

Perspective

Coupling climate-social system for effective climate governance

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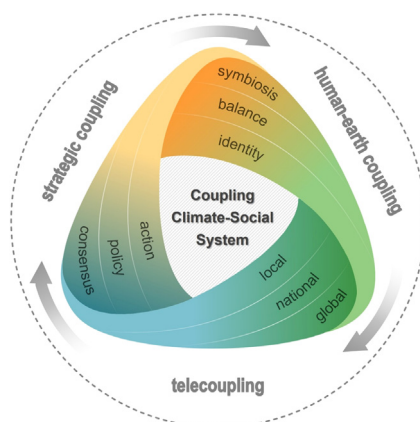
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HIGHLIGHTS

- Proposes the Möbius-based Coupling Climate-Social System (MCCSS) framework.
- Integrates three core principles for a multi-dimensional governance approach.
- Systematically bridges scales, values, and policies for coherent climate action.
- Advocates for a paradigmatic shift from reactive risk management to adaptive governance.

GRAPHICAL ABSTRACT



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ABSTRACT

Climate and social systems are intrinsically coupled. While this interconnectedness is widely recognized theoretically, operational frameworks in research and governance frequently compartmentalize them. This disconnection obscures critical feedback loops and impedes effective climate action. To address this, we introduce the Möbius-based Coupling Climate-Social System, which is structured around three core principles: human-Earth coupling, telecoupling, and strategic coupling. Effective climate governance requires the active integration of these three coupling dimensions. This entails fostering a balance between human activities and the environment, promoting cross-scale climate-social governance, and ensuring strategic coupling through the coordination of policies, planning, and actions.

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1. Introduction

Climate and social systems are intrinsically intertwined and co-evolutionary (Bastien-Olivera and Moore, 2022; Ellis, 2024). Historically, climate change has been driven by both natural and anthropogenic factors. However, in recent decades, human activities have emerged as the primary driver (Stone et al., 2009), with fossil fuel combustion and land-use changes fundamentally altering Earth's energy balance. The acceleration of atmospheric CO₂ persists (Lan et al., 2025), potentially driving Earth systems toward multiple tipping points (Steffen et al., 2015). Consequently, addressing these coupled dynamics has become a focal point in geographical and sustainability research (Fu et al., 2025).

Against this biophysical backdrop, current climate governance generally operates through two interconnected pillars: mitigation, targeting root causes through emission reductions, and adaptation, addressing impacts through localized resilience measures. However, these societal responses are shaped by distinct and complex political and economic forces (Liu et al., 2023). Regarding mitigation, the global shift toward renewable energy has been fundamentally accelerated by technological cost reductions and immediate socio-economic imperatives, rather than environmental awareness alone (Patt, 2017). Consequently, the persistent crisis is increasingly understood not as a simple collective action dilemma, but as a result of structural path dependencies—fundamentally rooted in “carbon lock-ins” (Unruh, 2000)—and deep-seated distributional conflicts between emission-intensive and climate-vulnerable interests (Aklin and Mildenberger, 2020). The ability of entrenched, high-emission interest groups to effectively block progress, even when a majority of the population favors stronger climate policy, exposes profound democratic deficits within current governance institutions. Furthermore, the operational boundaries between mitigation and adaptation are not strictly binary. Capacity constraints are a profound barrier to mitigation efforts as well; a lack of green finance and technical capabilities stymies decarbonization not only in developing countries but also across numerous municipalities in developed nations, particularly within shrinking urban areas confronting economic stagnation. Behavioral feedback also contributes, yielding an average annual mitigation effect of about 0.25 % for each 1 °C rise in temperature (Jarvis et al., 2012). In the domain of adaptation, social factors also amplify climate vulnerabilities. Poor governance weakens early warning systems, and wealth disparities shape adaptive capacity. By 2050, around 143 million people in the Global South will face climate migration risks (Rigaud et al., 2018). Gender inequality exacerbates disaster mortality, with women's death rates 14 times higher than men's (United Nations Office for Disaster Risk Reduction, 2022). To better understand these complex interactions, a large number of coupled climate-society models have emerged in recent years (Buhaug and von Uexkull, 2021; Gleick et al., 2018; Sang et al., 2025; Zhao et al., 2025). While these models represent a significant step forward, it remains crucial to strengthen the intrinsic connections between climate and social systems from a more holistic and systemic perspective to effectively address the deeply intertwined crises of climate and society.

Effectively navigating this complexity requires grounding the analysis in the foundational scholarship of Social-Ecological Systems (SES) and complex adaptive systems theory. The concept of SES fundamentally challenges the artificial delineation between social and natural systems, positing them instead as linked, co-evolving entities embedded within the biosphere (Berkes and Folke, 1998). Within this paradigm, humanity is not merely an external driver but an inhabitant of the biosphere, shaping and being shaped by its dynamics from local to global scales (Folke et al., 2016).

Understanding the dynamics of these coupled systems requires moving beyond equilibrium-centered views to embrace resilience. Originally defined by Holling (1973) as the capacity of a system to absorb disturbance and persist while retaining its same function, structure, and feedbacks, ecological resilience is distinct from engineering stability, which merely measures the speed of return to an equilibrium state. Resilience

thinking further integrates adaptability—the capacity of actors to influence resilience and sustain development within the current stability domain—and transformability—the capacity to create a fundamentally new system when ecological, economic, or social structures make the existing system untenable (Folke et al., 2010).

These non-linear dynamics are effectively captured by the “Panarchy” framework, which conceptualizes SES as nested sets of adaptive cycles moving through four distinct phases: exploitation (r), conservation (K), release (Ω), and reorganization (α) (Holling, 2001). This heuristic model elucidates how cross-scale interactions shape system trajectories: fast, small-scale cycles can trigger “revolt” processes that influence larger systems, while slower, large-scale systems provide “memory” that stabilizes and constrains smaller cycles (Gunderson and Holling, 2002). In the context of climate change, these cross-scale dynamics help visualize how individual perceptions and behaviors can cascade upward to influence broader community resilience (Greenlees and Cornelius, 2021).

The governance of such complex, multi-scalar systems necessitates moving beyond monocentric, top-down approaches. Polycentric systems—characterized by multiple governing authorities at differing scales—offer enhanced opportunities for experimentation, learning, and building trust, which are critical for navigating the deep-seated distributional conflicts and carbon-intensive path dependencies that obstruct effective climate mitigation (Aklin and Mildenberger, 2020; Roberts et al., 2025). This aligns with adaptive governance, which emphasizes social learning, network building, and the flexibility to navigate social-ecological transitions (Folke et al., 2005). Furthermore, when systems approach critical tipping points or exist in undesirable states (e.g., poverty traps or rigid carbon lock-ins), governance must shift from merely adaptive to transformative governance, actively triggering and steering regime shifts toward more sustainable trajectories (Chaffin et al., 2016).

While the theoretical insights from SES and resilience scholarship provide a robust ontological foundation for understanding climate-society interactions, a disconnect persists when these concepts are translated into operational models and policy frameworks. Significant advancements in Earth System Models (ESMs) and Integrated Assessment Models (IAMs) have vastly improved our ability to project physical climate changes and macroeconomic trajectories (Brauer et al., 2023; Keenan et al., 2023; Lou et al., 2025; Shen et al., 2025). However, these operational tools often face structural challenges in fully capturing the endogenous coupling logic emphasized in SES theory.

Specifically, mainstream modeling approaches typically conceptualize social dynamics—such as technological adoption, policy interventions, or behavioral shifts—as exogenous scenarios (e.g., SSPs) or linear responses to climate stress (Howard and Livermore, 2019; Ostrom, 2009). Such approaches tend to linearize complex adaptive systems, implicitly filtering out the non-linear “revolt” dynamics described in the Panarchy framework, where small-scale social innovations or local crises can cascade upward to trigger rapid systemic transformation. By treating critical societal feedbacks—such as institutional inertia, social tipping points, or inequitable adaptive capacity—as static boundary conditions rather than co-evolving variables (Beckage et al., 2022; Woodard et al., 2019), current assessments may underestimate the compounding nature of socio-climatic risks (Hahnel and Frings, 2024; Rising et al., 2022).

In the realm of governance, a similar gap exists between the theoretical ideal of polycentricity and the reality of policy implementation. While the IPCC AR6 highlights the intricate web linking climate action to Sustainable Development Goals (SDGs), existing governance structures often struggle to navigate the “wicked” nature of these linkages (Ogier et al., 2025). Decision-making systems frequently operate in silos, often resulting in structural conflicts rather than synergies within climate policy mixes (Wu et al., 2026). Consequently, these systems tend to optimize for specific sectors (e.g., flood control for adaptation or carbon pricing for mitigation) without the adaptive flexibility required to manage cross-scale interactions. Furthermore, when systems approach

resilience thresholds—where incremental adaptation is no longer sufficient—conventional governance frameworks often lack the mechanisms to facilitate deliberate transformative governance, risking entrapment in undesirable rigidity traps.

This systemic rigidity stems from the historical decoupling of the two pillars in policy practice. While mitigation is traditionally managed through compliance-driven mechanisms like NDCs and adaptation through localized NAPs, their operational barriers—such as capacity constraints and equity gaps—are increasingly recognized as porous. Despite these institutional silos, the two remain physically coupled—sectors like land use simultaneously dictate emission trajectories and systemic vulnerability. The primary challenge lies in the operational gap between these complex dynamics and fragmented governance structures. To bridge this divide, we propose the Möbius-based Coupling Climate-Social System (MCCSS) framework. Synthesizing insights from human-Earth coupling, telecoupling, and strategic coupling, the framework functions as a boundary object to reconcile the rigid compliance and political friction of mitigation with the context-specific flexibility of adaptation. Furthermore, we ground these concepts in complexity science methodologies to offer actionable, methodology-backed pathways for developing resilient climate strategies. Specifically, Multi-Regional Input-Output (MRIO) models and Social Network Analysis (SNA) can be employed to systematically trace telecoupled emission flows and explicitly reveal the specific nodes where high-emission interests are structurally entrenched, while Agent-Based Modeling (ABM) provides vital insights into complex, localized social feedbacks.

2. Imperative of climate and social coupling

2.1. Risks of ignoring human-Earth coupling

The process of climate governance inherently involves building resilience under conditions of high uncertainty. However, the core challenge has shifted from a theoretical trade-off between economic imperatives and ecosystem integrity to a political reality shaped by distributional conflicts and systemic rigidities. How societies navigate these tensions depends not merely on collective risk perceptions, but critically on the shifting power dynamics between entrenched emission-intensive incumbents and an emerging coalition of “green” interests, such as the renewable energy sector (Meckling et al., 2015). Although climate-vulnerable groups have been vocal participants since the inception of global climate negotiations, they continue to be marginalized by profound systemic power asymmetries within this structural conflict (Stoddard et al., 2021).

In the domain of mitigation, for instance, although 69 % of the global population supports a climate tax (at a 1 % income threshold), policy implementation is frequently derailed (Andre et al., 2024). While technical and financial capacity constraints undeniably impede progress—especially in developing nations and resource-strapped municipalities—the persistent failure to enact such widely supported policies highlights how targeted political obstruction by emission-intensive groups actively widens the “intention-action gap” (William et al., 2020). Meanwhile, in the context of adaptation, social stratification—including economic status, gender, and ethnicity—plays a major role in shaping unequal climate risks (Diep and McPhearson, 2022). Climate hazards severely undermine livelihood security for women in Sub-Saharan Africa and impair the long-term well-being of Indonesian girls born during disaster years (Burke et al., 2015; Maccini and Yang, 2009). These cases reveal that physical risks are inextricably coupled with social structural inequities.

Crucially, while earlier scholarship often attributed governance failures to cognitive biases or bounded rationality (Boulding, 2013; Lindahl et al., 2024), recent assessments suggest that these psychological factors are secondary to structural path dependencies. The failure to bend the global emissions curve is increasingly understood as a consequence of deliberate obstruction by vested interests and the

path-dependent stability of fossil-fuel-based socio-economic systems (Roberts et al., 2025; Stoddard et al., 2021). Mainstream climate-economic models that rely on simplified assumptions of human behavior often fail to capture these non-linear political feedbacks (Gilligan, 2018; Kugler et al., 2008; McNamara and Keeler, 2013). Neglecting this complexity within human-Earth coupling leads to overly optimistic assessments of transition pathways, as it underestimates the system’s resistance to change driven by incumbent power structures (Colgan et al., 2020).

While foundational SES research has long established the intrinsic interconnectedness of human and natural domains (Ostrom, 2009), the proposed MCCSS framework advances this legacy by addressing the specific limits of linear coupling in governance contexts. In contrast to traditional SES perspectives that often conceptualize these interactions as binary linkages, MCCSS conceptualizes them as a topological continuity, offering a more precise explanation for the “intention-capacity divide” mentioned above. Regarding the paradox where high public support for mitigation fails to translate into policy, MCCSS reinterprets this not as a mere implementation failure, but as a topological “twist”—where a positive social consensus is inverted into a negative governance outcome upon traversing the rigid manifold of interest-based politics. Similarly, regarding adaptation in sub-Saharan Africa, MCCSS maps how physical hazards are continuously reshaped by social stratification into entrenched structural lock-ins. Thus, MCCSS provides a diagnostic tool for these non-linear phase transitions, enabling governance to anticipate how benign policy intentions can be distorted by distributional conflicts and structural inequalities.

2.2. Multi-scale climate governance dilemma

The vertical integration of climate governance encounters substantive tensions best understood through the lens of Panarchy theory and the challenge of disentangling distinct scale approaches (Gunderson and Holling, 2002; Termeeer et al., 2010). Cross-scale dynamics oscillate between “remember” (stabilization driven by large, slow cycles) and “revolt” (disruption from smaller, fast cycles). Reevaluating this through a political economy perspective, the dominant large-scale, slow-cycle system is not an abstract climate equilibrium, but the deeply entrenched global fossil-fuel economy. Consequently, frameworks like the UN-FCCC and the Paris Agreement are more accurately conceptualized as relatively small-scale, fast-cycle institutional attempts to instigate a “revolt”—aiming to trigger systemic reorganization toward decarbonization. However, these progressive “revolts” are continuously constrained by the massive “remember” function of the fossil-dependent macroeconomic structure, which actively resists disruption. The institutional shift from the Kyoto Protocol—whose self-determined targets were embedded in a rigid, legally binding compliance architecture (Depledge, 2022)—to the flexible, non-binding pledge-and-review system of the *Paris Agreement* reflects an attempt to manage this tension. Yet, despite these varied governance configurations, both frameworks struggle to overcome the gravitational pull of fossil-capital inertia between nations.

This dialectical friction is starkly evident in regional mitigation frameworks, revealing the nested and complex nature of these dynamics. The European Union’s “Fit for 55” package, targeting a 55 % emission reduction by 2030 through ETS reforms and a proposed ban on internal combustion engine (ICE) vehicles (Ovaere and Proost, 2022), acts as a macro-regional “revolt” against global carbon lock-in. Yet, when translated into top-down mandates within the EU subsystem, it triggers a counter-“revolt” from member states seeking to protect local economic resilience and established industrial paths. For example, Italy has advocated for delaying the planned 2035 ICE ban, citing concerns regarding the resilience of its automotive sector—an industry that accounts for 5.2 % of the nation’s GDP (International Trade Administration, 2022). Meanwhile, Czechia and six other member states have resisted stringent emission penalties (specified at €95 per g/km above

the target limit, referenced against a 94 g/km benchmark for 2025) (Włodarczak-Semczuk, 2024). These divergent stances extend beyond mere compliance failures; they illustrate a fractal Panarchy where local resistance serves the “remember” function of national fossil-industrial paths against regional climate disruptions. As noted in studies of managed retreat, such frictions reflect the complex convergence of psychological perspectives and adaptive systems theory, where local resistance is an inherent property of the system’s adaptive cycle (Greenlees and Cornelius, 2021).

Challenges also arise in horizontal coordination across jurisdictions, reflecting the complexities of polycentric governance (Ostrom, 2010). Independent decision-making centers are shaped by distinct institutional path dependencies and divergent problem framings. Even progressive environmental policies face political reversals driven by interest group obstruction. Notable examples include the volatility in US climate policy. While the Inflation Reduction Act (IRA) represented a significant institutional “revolt” aimed at disrupting domestic carbon lock-in, it remains a relatively fast and vulnerable policy cycle compared to the overarching fossil-fuel economic base. From a revised Panarchy perspective, the IRA’s vulnerability to being dismantled under a renewed Trump administration (Shah, 2025) is not a failure of climate governance’s institutional memory; rather, it vividly demonstrates the formidable power of the incumbent fossil-fuel system’s “remember” function. This macro-structure forcefully snaps back to conserve (K phase) its original, emission-intensive stability domain, actively expelling disruptive policies.

Similarly, mechanisms like the EU’s Carbon Border Adjustment Mechanism (CBAM) and China’s “Dual Carbon” strategy illustrate the tension between domestic goals and global coordination. CBAM aims to mitigate carbon leakage, yet its unilateral implementation raises concerns about transferring carbon costs to developing nations, highlighting the distributional conflicts inherent in global trade-climate coupling (Döbbling-Hildebrandt et al., 2024; Fedoseeva and Zeidan, 2018; Steinebach et al., 2021). Furthermore, as international trade and supply chains deepen, the institutional and technological rigidities of fossil-fuel-dependent economies are increasingly exported, leading to a phenomenon of “globalizing carbon lock-in” (Unruh and Carrillo-Hermosilla, 2006). Dominant national policy pathways often oversimplify these complex responses. In particular, insufficient consideration of how these entrenched lock-ins propagate through telecoupling substantially increases the costs of coordination.

Impediments to effective climate action also manifest at individual and behavioral scales. For instance, while Shanghai’s waste sorting program reduced daily carbon emissions from the municipal waste system by 31.83 % (Wang et al., 2021), regulatory fragmentation and asymmetric incentives constrained community participation (Zhang et al., 2023). Similarly, Paris’s “Paris Respire” initiative relies heavily on context-specific socio-cultural factors. These examples indicate that multi-scale governance frameworks frequently treat climate change primarily as an external environmental problem, potentially overlooking critical socio-behavioral dimensions. Consequently, significant disconnects persist in integrating climate considerations vertically (global-to-individual) and horizontally. The absence of a unifying perspective that organically couples climate physical processes with multi-scale governance models limits the capacity to drive systemic transformation. To bridge this gap, governance must align high-level “remember” functions with the granular, context-specific variables of local resource systems (Ostrom, 2007).

2.3. Rethinking the boundaries of mitigation and adaptation governance

The shortcomings of current governance modes are best understood through the dual lenses of political economy and resilience theory (Folke et al., 2010). Effective global climate governance requires stable and sustained strategic coupling (Galaz et al., 2011); yet, as evidenced by the outcomes of the recently concluded COP30 in Belém, governance

systems remain trapped in deep-seated trajectories that hinder the transformative action required to navigate the Anthropocene.

In the realm of mitigation, the primary barrier is not merely collective action dilemmas or stochastic volatility, but rather entrenched path dependencies and distributional conflicts driven by emission-intensive interests. This obstruction is starkly evident in the policy reversals of major powers. While early U.S. programs struggled to keep pace with risks (Shi and Moser, 2021), the confirmed withdrawal of the United States from the *Paris Agreement* in 2025 introduces significant stochasticity to global cooperation (Trump, 2025). Viewed through Holling’s (1973, 2001) adaptive cycle, this is not just a chaotic “release” (Ω) phase, but a manifestation of systemic rigidity where vested interests force a reversion to fossil-fuel-dependent “conservation” (K) phase, preventing beneficial reorganization.

While sharing these pervasive capacity limitations, adaptation governance faces a more localized set of structural impediments centered on capacity gaps rather than compliance failures. Despite ostensibly inclusive measures, vulnerable communities remain barred from resources by social stigma or economic rigidities (Roth, 2018). Technological solutions often inadvertently reinforce these infrastructure lock-ins through design biases (Gaskin et al., 2017), rather than fostering the diverse pathways necessary for resilience (Biggs et al., 2012). At the international level, this “adaptation gap” is widening. Developing countries now face an adaptation finance demand 12–14 times greater than existing inputs, yet international public adaptation finance paradoxically declined by 7 % from 2022 to 2023 (UNEP, 2025a).

This persistent fragmentation culminates in a fractured global landscape. On one hand, mitigation efforts are stalled by insufficient ambition and the challenge of enforcing compliance with NDCs—where aggregate efforts still imply a warming of 2.6–2.8 °C (UNEP, 2025b). On the other hand, adaptation efforts are undermined by a lack of domestic legislative frameworks and financial support. While the “Baku to Belém” roadmap aims to mobilize \$1.3 trillion (Jiang et al., 2026), systemic fragility threatens outcomes. Although mitigation heavily contends with enforcement challenges and political resistance, a defining barrier for vulnerable states regarding adaptation is often a “legislative void”. For instance, rather than a failure to comply with international rules, countries like Ethiopia face internal hurdles in establishing dedicated legal architectures to address extreme weather impacts, trapping them within a cycle of diminished adaptive capacity (Bisare Bitire, 2022). In essence, without addressing these interconnected structural feedback loops—recognizing that political friction and capacity deficits concurrently impede both mitigation and adaptation—current frameworks cannot bridge the widening “commitment-action gap”.

3. Coupling climate-social system as a Möbius strip

3.1. Möbius and the coupling climate-social system

We propose the MCCSS framework, an innovative approach to conceptualizing climate-social interactions. Drawing on the topology of the Möbius strip—characterized by its single continuous surface—the framework metaphorically represents the dynamic feedbacks between climate and social systems as an inseparable whole, transforming previously distinct entities (Fig. 1). Addressing the limitations of traditional linear paradigms, which often separate spatial-temporal scales and system elements, the MCCSS framework utilizes three key topological properties (Pickover, 2005; Wunderlich, 1962). One is monosurface continuity: the 540° twist merges distinct surfaces into a single whole. This dissolves the “Nature-Society” binary, symbolizing that “externalities” (e.g., carbon emissions) are topologically destined to cycle back as “internal risks” (e.g., local floods), forcing a unification of mitigation causes and adaptation consequences. Another is connectivity: any two points are reachable along the surface. This implies that influences propagate without crossing artificial borders, challenging the “siloes” governance of adaptation where vulnerable commu-

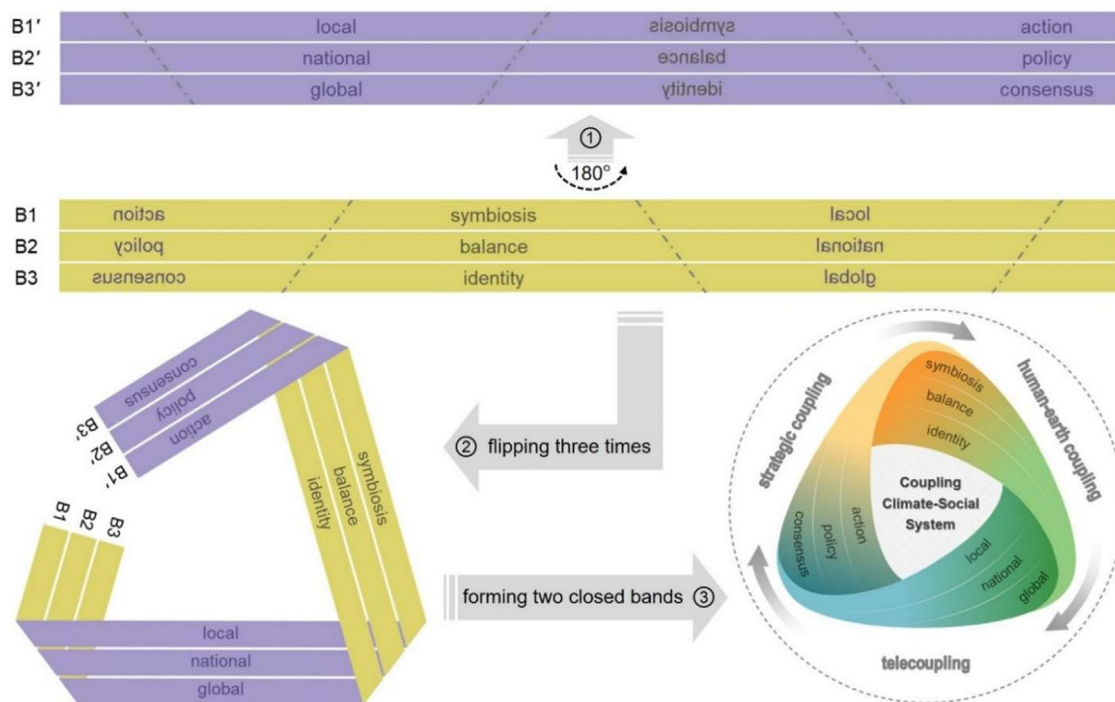


Fig. 1. Möbius-based Coupling Climate-Social System. (1) B1 represents the front of band 1, and B1' represents the back of band 1 after rotating horizontally by 180°; the meanings for the other two bands are similar. (2) After three flips, the three bands appeared with the front and back sides connected as B1-B3', B2-B2', and B3-B1'. (3) The three bands are joined together to form two closed bands, and the original bands 1 and 3 are merged into a closed band twice the length of band 2.

nities are often treated as isolated victims rather than interconnected nodes. The last is non-orientability: the inherent twist means local orientation reverses upon traversal. This challenges the notion of linear progress, visualizing how global mitigation consensus can be “inverted” into local policy failure due to the structural friction of conflicting interests.

This topological conceptualization offers a distinct yet complementary lens to the Panarchy framework (Gunderson and Holling, 2002). While Panarchy models complex systems through temporal phase transitions (exploitation, conservation, release, reorganization), MCCSS visualizes them through spatial geometry to address institutional path dependencies. The Möbius “twist” acts as a structural analog to the “release” phase, but explicitly highlights the ontological inversion of variables. For instance, it maps where the physical accumulation of CO₂ (a linear process) “twists” into chaotic social crises (a non-linear response). Furthermore, whereas Panarchy analyzes interactions between nested scales via “revolt” and “remember”, MCCSS unifies these scales onto a continuous manifold. This allows for mapping emergent properties not just as products of cross-scale friction, but as phenomena propagating along a seamless surface where the distinction between “global” drivers and “local” responses dissolves—a critical perspective for understanding the telecoupled nature of climate justice.

Crucially, the structural implications of the Möbius transformation reconfigure the logic of mitigation and adaptation governance. We apply a tripartite division to the pre-transformation plane: outer (Global-Consensus), middle (National-Policy), and inner (Local-Action) bands. Following the Möbius transformation, a dual topological transition occurs: First, the “outer” Global and “inner” Local bands merge into a single, continuous loop, embodying the Global-Local Continuum of Adaptation Logic; this reflects how global physical changes translate directly into local impacts, necessitating a governance logic of seamless connectivity and human-Earth coupling that transcends borders. Conversely, the “middle” National band forms a distinct, self-contained loop representing the National Fulcrum of Mitigation Logic—a structural isolation that identifies the sovereign bottleneck where distributional conflicts

and path dependencies are most acute. Consequently, the framework demonstrates that effective governance is non-uniform: adaptation requires traversing the continuous Global-Local loop (via telecoupling), while mitigation depends on unlocking the self-contained National loop (via strategic coupling) to prevent it from becoming a “rigidity trap”. This geometric rationale explains why relying solely on global consensus fails if it does not explicitly engage with the political economy of the national middle band.

3.2. Three principles of MCCSS

The MCCSS framework operationalizes the topological logic described above via three functionally complementary coupling principles. These principles are designed to address the specific governance failures identified in Section 2, distinguishing between the structural needs of mitigation and adaptation. Telecoupling integrates global and local elements to trace cross-boundary climate impacts (Hull and Liu, 2018; Liu et al., 2013). In traditional analyses, this serves as a tool to map flows like embodied carbon (Liu et al., 2016). However, within MCCSS, we reframe telecoupling as the constitutive topological substrate that enforces “connectivity”. Its primary governance function is to reveal the hidden distributional conflicts inherent in climate mitigation. By establishing that any two points are reachable along a continuous surface, telecoupling ontologically transforms distant “externalities” into “internal” feedback loops, preventing high-emission regions from displacing the costs of their growth. The EU’s CBAM exemplifies this dynamic. Conventionally viewed as a trade border measure, the MCCSS perspective reveals CBAM as a mechanism attempting to manage telecoupled friction. Since the consumer in Berlin and the factory in Asia exist on the same continuous surface, CBAM acts as a governance feedback loop trying to internalize “distant” emissions back into local cost structures. By visualizing these connections, telecoupling exposes that the “border” is an artificial construct obstructing effective mitigation, compelling governance to treat global supply chains as a unified field of action where carbon responsibilities cannot be offshored.

Strategic coupling focuses on relational elements, i.e., consensus, policy, and action, to build institutional coherence necessary to overcome the political obstruction and path dependencies that characterize national-level governance. It aims for synergy horizontally (with human-Earth coupling) and supports vertical scale transitions (inherent in telecoupling). Notably, this principle complements yet distinguishes itself from adaptive governance (Folke et al., 2005). While adaptive governance prioritizes flexibility to manage uncertainty, it often lacks the structural power to break entrenched rigidities; therefore, strategic coupling acts as a necessary “binding” mechanism. Functionally, it operates through dual logic: for mitigation, it navigates the political economy of transition by aligning global “compliance” architectures with rigid national industrial policies to unlock carbon lock-ins; for adaptation, it bridges “legislative voids” in vulnerable states by integrating fragmented local experiments into national planning codes. This dynamic is exemplified in urban flood resilience, where strategic coupling integrates isolated innovations—such as rain gardens—into broader regulatory frameworks and global green finance standards, effectively resolving the fragmentation between the “revolt” of local action and the “remember” of national regulation.

Human-Earth coupling emphasizes value-oriented elements, such as symbiosis, balance, and identity, and serves as the primary operational logic for adaptation and resilience. Crucially, this concept extends beyond the functionalist logic of social-ecological fit (Epstein et al., 2015), which aligns institutional boundaries with biophysical scales, or the resource-centric perspective of adaptive capacity (Siders, 2019), which focuses on economic or technological assets. While these traditional concepts address the mechanisms and means of adaptation, human-Earth coupling targets the deeper axiological dimension—the motivations and meanings that drive or inhibit action. It seeks to mediate tensions between value rationality and instrumental rationality by analyzing how risk perception and social norms shape the topological “twist” of governance outcomes, ensuring that policy design accounts for the non-linear interplay between physical risk and social values. The practical necessity of this distinction is starkly illustrated by the case of “Managed Retreat” in coastal zones. From a social-ecological fit perspective, the solution is structural (aligning zoning laws with sea-level rise); from an adaptive capacity perspective, it is economic (providing funding for relocation). However, policy failure often persists despite these measures due to human-Earth coupling factors, specifically the deep psychological attachment to “place” and local identity. Residents may reject economically rational buyout offers because their identity is inextricably “coupled” with the landscape. Consequently, the MCCSS framework underscores that effective climate governance requires integrating these individual cognitive frameworks and community meaning-making into core policy design, promoting a paradigm shift from technical solutions towards a relational restructuring of the human-nature interface.

The necessity for integrating these three coupling principles—telecoupling, strategic coupling, and human-Earth coupling—lies in their ability to resolve governance fragmentation by bridging the structural and axiological gaps inherent in traditional frameworks. Current models often neglect cross-scale interactions, but the MCCSS framework’s closed-loop configuration synthesizes these discrete aspects into a cohesive whole. Taking the CBAM as a paradigmatic example, effective implementation requires not only telecoupling to reveal the topological connectivity of embodied carbon (reframing “externalities” as internal feedbacks), but also strategic coupling to structurally align international trade rules with national policies (preventing fragmentation), and a foundation of human-Earth coupling to secure the axiological consensus necessary for industrial transition. While telecoupling identifies the spatial connectivity of the problem, MCCSS proposes a holistic solution by integrating it with strategic principles that build resilient institutional architectures and human-Earth principles that mediate deep-seated tensions between resource management and cultural identity.

In summary, the MCCSS framework utilizes a topological metaphor to reframe the climate-social system not as a collection of binary dualisms, but as a dynamic, evolving, and non-orientable continuum. Theoretically, it moves beyond mechanistic reductionism, offering a geometric logic suited to the emergent and adaptive characteristics of the Anthropocene. Practically, its structure provides an integrated toolkit for cross-scale governance, where mitigation requires conquering distance (telecoupling) and resistance (strategic coupling), while adaptation requires bridging the gap between physical survival and human meaning (human-Earth coupling). Paradigmatically, this signifies a crucial transition from reactive crisis management towards fostering resilient coexistence, promoting a systemic and ontological restructuring of human-Earth relations amid accelerating climate change.

4. Towards an effective climate governance

4.1. Reconfiguring climate governance cognition

Cultivating effective governance requires a fundamental cognitive shift from reactive crisis management to proactive structural diagnostics (Table 1). Through the MCCSS perspective, this entails recognizing that climate failures are not merely technical errors but topological inevitabilities when governance ignores the continuous feedback between human and Earth systems. Consequently, governance cognition must transition from assuming discrete system boundaries to managing structural coupling: establishing accountability through telecoupling (tracing carbon flows), stabilizing transitions through strategic coupling (managing political friction), and ensuring equity through human-Earth coupling (aligning values). This shift necessitates a new methodological approach: topological Monitoring. Unlike traditional linear policy cycles (e.g., plan-do-check-act) that monitor variables within a stable state, topological monitoring tracks continuous trajectories to identify when the system approaches the Möbius “twist”—the critical threshold where benign inputs are ontologically inverted into malignant outputs.

Crucially, this monitoring must distinguish between the specific structural pathologies of mitigation and adaptation. For mitigation, governance must monitor the “political twist”—identifying the specific moment where a global consensus (e.g., an NDC pledge) enters the national political manifold and is threatened to be inverted by distributional conflicts and obstruction into policy failure. Conversely, for adaptation, monitoring focuses on the “vulnerability twist”—tracking where physical climate hazards intersect with social stratification and legislative voids, threatening to invert a manageable weather event into a humanitarian disaster. This differentiation moves governance beyond generic “risk management” to a precise diagnosis of whether the failure stems from political resistance (in mitigation) or institutional deficits (in adaptation).

To reveal leverage points within this continuum, governance must identify the “minimal cyclic period” along the system’s central axis—conceptually aligning with the “release” and “reorganization” phases of the adaptive cycle. Operationalizing this means identifying windows of political opportunity to intervene while the system is malleable. In mitigation, this involves pinpointing moments of systemic instability (e.g., energy price shocks or geopolitical shifts) to break carbon lock-ins and rewrite path dependencies, rather than fighting against rigid structures during stability phases. Even before the escalation of the climate crisis, substantial efforts were underway to advance the research, development, and deployment of energy-efficient and low-carbon technologies, alongside initiatives to cultivate public awareness and support (Unruh, 2002). Such proactive groundwork ensures that when a window of political opportunity arises, society possesses the requisite capacity to disrupt established path dependencies. In adaptation, it implies utilizing post-disaster recovery phases not merely to return to the status quo, but to fundamentally fill “legislative voids” and address inequality before rigid structures solidify again. Furthermore, instead of assuming uniform global effects, this approach requires mapping topo-

Table 1
Mapping governance failures to Möbius-based Coupling Climate-Social System (MCCSS) topological solutions.

Governance Domain	Key Governance Failure	Topological Structure	Coupling Principle	Operational Toolkit & Mechanism
Mitigation	Distributional conflict & obstruction	The National fulcrum Self-contained middle loop	Strategic coupling & telecoupling Focus: Binding & Accountability	Diagnostic: MRIO & SNA Governance: Binding architectures (e.g., Climate Clubs, CBAM)
Adaptation	Legislative voids & capacity deficits	The global-local continuum Merged outer/inner loops	Human-Earth coupling Focus: Values	Diagnostic: ABM Governance: Transmission mechanisms linking global finance directly to local planning codes

logical heterogeneity, analyzing how local disturbances propagate along the non-orientable manifold to generate global asymmetries and distinct “tele-connected” pathways of risk.

4.2. Effective climate action via MCCSS

To operationalize these cognitive shifts, we first propose a robust methodological toolkit to diagnose specific governance failures (Ye et al., 2026). For mitigation, we suggest utilizing MRIO models and SNA not merely to trace connectivity, but to visualize the hidden distributional conflicts of embodied carbon. Specifically, MRIO models can trace embodied carbon flows across complex global supply chains to unmask spatial inequities, while SNA can map the relational power structures and lobbying networks of the actors involved. By mapping the “tele-connected” nodes that act as barriers in the global network, these tools operationalize telecoupling as a measurable substrate, revealing where high-emission interests are structurally entrenched. The empirical necessity of pinpointing these specific nodes is powerfully underscored by the Carbon Majors database (Heede, 2013), which traces the vast majority of cumulative global greenhouse gas emissions to a remarkably small number of corporate and state entities. For adaptation, we recommend ABM to simulate how heterogeneous values and social stratification emerge into collective behaviors. This captures the non-linear “human factor” of human-Earth coupling, allowing policymakers to anticipate where “legislative voids” or inequality might invert a physical hazard into a social crisis.

Second, governance institutions must be restructured to provide structural binding against political volatility. Building on adaptive governance, which emphasizes flexibility (Chaffin et al., 2016; Folke et al., 2005), MCCSS adds the necessary layer of strategic coupling to prevent fragmentation. In the realm of mitigation, recognizing that the international climate behavior of states is inextricably tied to the balance of power among domestic interest groups (Aklin and Mildenberger, 2020), simply attempting to impose top-down “binding architectures” has historically proven insufficient. Instead, strategic coupling must actively undermine the blocking power of emission-intensive incumbents (Green, 2021). To operationalize this, international frameworks should pivot toward “catalytic cooperation” (Hale, 2020). Because many mitigation actions yield both public and private benefits, the primary challenge is overcoming the initial activation barrier. By helping progressive actors to start up and scale up, international institutions can trigger self-reinforcing feedbacks that progressively shift domestic power balances, ultimately creating the political conditions where robust, binding commitments become viable. Conversely, in adaptation, it involves creating “transmission mechanisms” that connect global green finance standards directly with local urban planning codes, thereby bridging the capacity gap in vulnerable states. Furthermore, decision-making processes must address power dynamics through Power-Sensitive Scenario Planning. Unlike generic participatory approaches, this method explicitly negotiates the trade-offs between incumbent emission-intensive interests and climate-vulnerable communities, managing the “revolt” dynamics of local actors.

In summary, the MCCSS framework utilizes a topological metaphor to reframe the climate-social system not as a collection of binary du-

alisms, but as a dynamic, evolving, and non-orientable continuum. Theoretically, it moves beyond mechanistic reductionism, offering a geometric logic suited to the emergent and adaptive characteristics of the Anthropocene. Practically, its structure provides an integrated toolkit for cross-scale governance: reimagining telecoupling to map the distributional conflicts of mitigation; employing strategic coupling through catalytic cooperation to overcome incumbent political obstruction; and harnessing human-Earth coupling to bridge the equity gaps in adaptation. Looking forward, future research must prioritize topology-aware analyses that rigorously account for real-world constraints, including geopolitical volatility and economic fragmentation. Ultimately, adopting MCCSS signifies a paradigmatic transition from reactive crisis management towards a systemic restructuring of human-Earth relations essential for resilient coexistence.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRediT authorship contribution statement

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